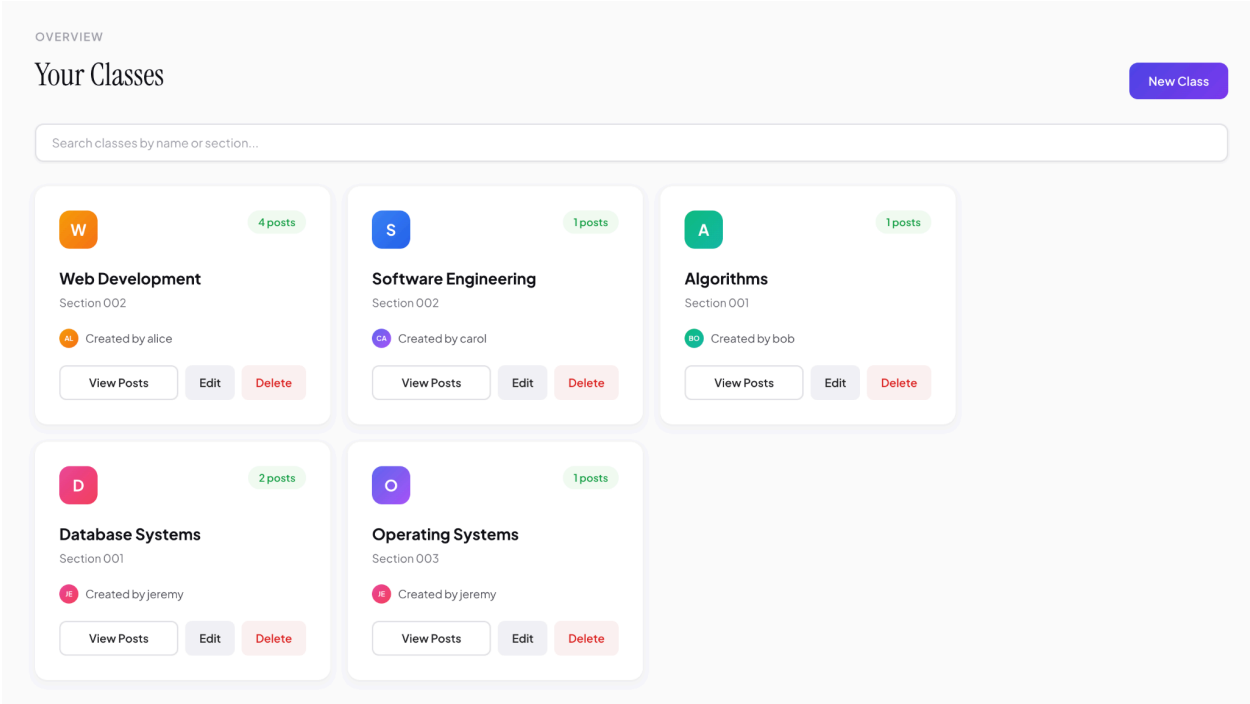
5. Reporting

Report 1: Get all classes

To get the data for the "Your Classes" page, the database queries the accounts, posts, and classes tables to display class names, who created them, and how many posts they have. This is useful to users because it shows them what classes already exist and are joinable.

```
SELECT
  c.*,
  a.name as creator_name,
  COUNT(p.id)::int as post_count
FROM classes c
JOIN accounts a ON c.creator_id = a.id
LEFT JOIN posts p ON p.class_id = c.id
GROUP BY c.id, a.name
ORDER BY c.created_at DESC
```

Fig. 20: Report for all classes



Report 2: Fetch posts with skill-match ranking

When a user loads the posts feed, posts are ranked by how many skills the group's current members share with the logged-in user, so the most relevant groups appear first.

```
const query = `
    WITH user_skills AS (
      SELECT skill_id FROM account_skills WHERE account_id = $1
    )
    SELECT
      p.*,
      a.name as author_name,
      c.name as class_name,
      c.section as class_section,
      g.id as group_id,
      g.group_name,
      g.max_members as group_max_members,
      g.created_by as group_created_by,
      COALESCE((
        SELECT COUNT(*)::int
        FROM account_groups ag2
        JOIN account_skills ags2 ON ag2.account_id = ags2.account_id
        JOIN user_skills us ON ags2.skill_id = us.skill_id
        WHERE ag2.group_id = g.id
        AND ag2.account_id != $1
        AND ag2.is_pending = false
      ), 0) as skill_match_score
    FROM posts p
    JOIN accounts a ON p.author_id = a.id
    JOIN classes c ON p.class_id = c.id
    LEFT JOIN groups g ON p.group_id = g.id
    ${class_id ? "WHERE p.class_id = $2" : ""}
    ORDER BY skill_match_score DESC, p.created_at DESC
  `;
```

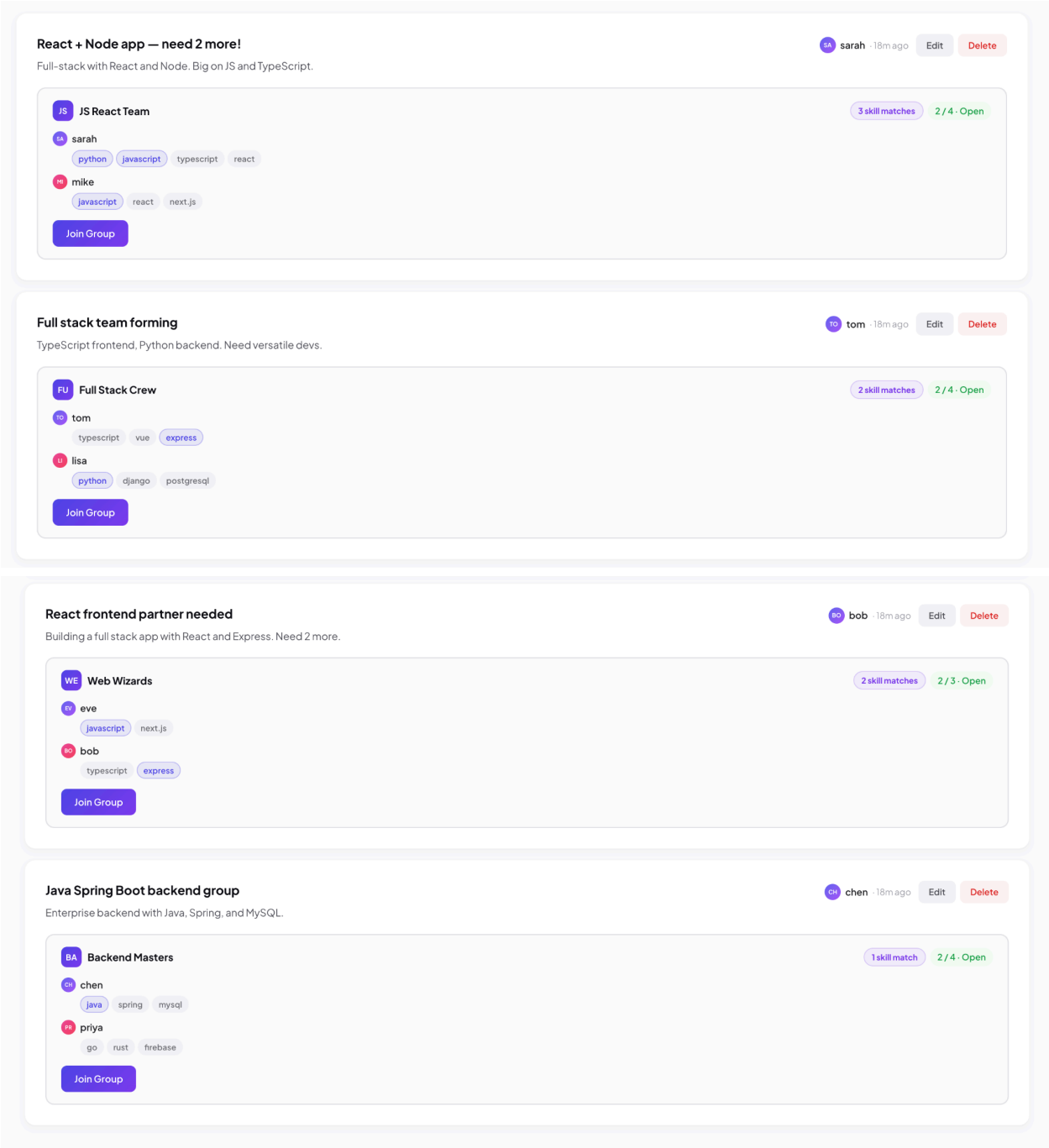



Fig. 21: Report for posts